

# INDIAN FMCG RETAIL SALES ANALYSIS

## Power BI Project – Executive Summary Report

### 1. Project Overview

This Power BI project analyzes Indian FMCG retail sales performance using transaction-level sales data. I built the dashboard to understand revenue, profit, orders, units sold, margins, customer characteristics, channels, store formats, brands, categories and city-level performance.

### 2. Key Performance Indicators

| KPI                       | Value  |
|---------------------------|--------|
| Total Revenue             | 39.34M |
| Total Profit              | 7.87M  |
| Total Orders              | 100K   |
| Total Units Sold          | 300K   |
| Profit Margin             | 20.02% |
| Average Order Value (AOV) | 393.60 |

### 3. Dashboard Pages & Analysis

- Executive Dashboard – A high-level business view using KPI cards and major sales/profit visuals.
- Sales Performance – Revenue and profit analysis across categories, channels, store formats, brands and cities.
- Customer Analysis – Customer-focused analysis using available customer attributes such as age, gender and loyalty.
- Interactive Analysis – A dedicated page with interactive slicers for exploring the report.

### 4. Important Business Metrics

- Revenue represents the total sales value generated by the business.
- Profit represents the earnings after considering the relevant cost.
- Profit Margin shows profit as a percentage of revenue.
- Average Order Value (AOV) indicates the average revenue generated per order.

### 5. Analysis Areas Covered

- Monthly Revenue Trend – tracks how revenue changes over time.
- Revenue and Profit by Category – compares sales contribution and profitability across categories.
- Sales and Profit by Channel – compares performance across available sales channels.
- Sales by Store Format – compares different store formats.
- Top Brands – highlights strong-performing brands based on revenue.

- City Performance – compares revenue across cities.
- Customer Analysis – uses available customer attributes for segmentation.
- Interactive Slicers – allow users to filter and explore the dashboard dynamically.

## 6. Data Preparation

- The dataset was checked in Power Query before visualization.
- Column quality and data types were reviewed.
- Customer Age contained missing values; these were not blindly replaced because the missing share was high.
- A dedicated DimDate table was created using the minimum and maximum Invoice\_Date values.
- Because Invoice\_Date contained date and time, an Invoice Date Only column was created for the Date-table relationship.

## 7. Key Takeaways

- The dashboard reports approximately 39.34M in total revenue and 7.87M in total profit.
- The overall profit margin is approximately 20.02%.
- The business processed approximately 100K orders and sold approximately 300K units.
- An AOV of 393.60 provides a useful benchmark for order-level sales performance.
- The report allows comparison of categories, channels, store formats, brands and cities to identify stronger and weaker areas.
- Interactive filtering makes the report useful for both high-level review and detailed analysis.

## 8. Conclusion

- This Power BI dashboard provides an end-to-end view of FMCG retail sales performance. It combines data preparation, a dedicated date model, DAX-based KPIs, time analysis, category/channel analysis, customer insights and interactive filtering in one reporting solution.
- The project demonstrates practical skills in Power BI, Power Query, DAX, data modeling, visualization and business-oriented analysis.

## 9. Tools & Skills Used

- Power BI • Power Query • DAX • Data Cleaning • Data Modeling • Date Table • KPI Design • Data Visualization • Interactive Slicers • Business Analysis